



MTH104 Quiz#3 7 July2023 Finals by MISS Mehwish

Virtual University Of pakistan (Virtual University of Pakistan)



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الرَّحِيمِ الرَّحْمَنِ اللَّهُ بِسْمِ

1. Let $S(A)$ denote the collection of all initial segments of elements in a well-ordered set A , and let $S(A)$ be ordered by set inclusion. Then A is similar to $S(A)$ and, in particular, the function $f : A \rightarrow S(A)$ defined by $f(x) = s(x)$ is a similarity mapping between A and $S(A)$.

Options:

- a) True
- b) False

Correct option: a) True

2. Let C be a non-empty set of non-empty sets. A function $f : C \rightarrow UC$ is called a choice function for C if $f(A) \in A$ for all $A \in C$.

Options:

- a) True
- b) False

Correct option: b) False

3. The ordinal number of set $\{a, \{b, c\}\}$ is _____.

Options:

- a) 0
- b) 1
- c) 2
- d) 3

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Correct option: d) 3

4. A is bounded if _____.

Options:

- a) A has only a lower bound.
- b) None of these.
- c) A has only an upper bound.
- d) A has an upper and lower bound.

Correct option: d) A has an upper and lower bound.

5. The ordinal number of the set P of counting numbers is denoted by ω .

Options:

- a) True
- b) False

Correct option: a) True

6. Suppose A and B are well-ordered sets, and suppose A is similar to an initial segment of B. Then A is said to be _____ than B.

Options:

- a) Shorter
- b) Longer

Correct option: a) Shorter

7. If an upper bound of A succeeds every other upper bound of A, then it is called the supremum of A.

Options:

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a) False

b) True

Correct option: b) True

8. Let S be a partially ordered set, and let A be a subset of S . An element n in S is called a lower bound of A if n precedes every element of A , that is, for every $y \in A$, we have $y \geq n$.

Options:

a) False

b) True

Correct option: b) True

9. Suppose A and B are well-ordered sets, and suppose A is similar to an initial segment of B . Then B is said to be _____ than A .

Options:

a) Longer

b) Shorter

Correct option: a) Longer

10. Let C be the set of countries on earth, thinking of each country as a collection of cities. Then UC is the set of all cities on earth, and the function f that assigns to each country its capital city is a choice function for C .

Options:

a) True

b) False

Correct option: b) False

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